OFA8 / OFA10 / OFA11 — Python `psm` ↔ MATLAB Validation Comparison

Full three-sample validation of the Python/Streamlit `psm` pipeline against the PhD student's MATLAB reference (`PDF_Matlab_updated.m`). Reference values and per-sample thresholds taken from his results deck (`OFA__1_.pptx`). Supersedes the OFA8-only validation summary.

Common settings (all three samples): scale-bar calibration 0.0365 µm/px (30 µm / 825 px, one imaging session, 4000×, 20 kV, det vCD); watershed ON with regional-maxima seeding; 8-connectivity; `bwareaopen(15)` equivalent; `lognfit` convention (ddof=1). Each sample segmented with **its own** intensity bands (below).

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1. Reference thresholds (his, per sample)

Sample	Pore	NiO	BZCY
OFA8	0–100	101–174	175–255
OFA10	0–75	76–180	181–255
OFA11	0–59	60–190	191–255

The bands shift substantially across samples (pore ceiling 100 → 75 → 59; NiO/BZCY split 174 → 180 → 190) because SEM brightness varies between samples. His own threshold sets are direct evidence for the **"no universal threshold"** limitation — cite them in the Methods/Limitations sections. Using OFA8's bands on OFA10/OFA11 produces systematically wrong volume fractions (pore too high, NiO too low, BZCY too high); matching his per-sample bands removes the discrepancy.

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2. Validation comparison — app vs his MATLAB

OFA8 (bands 0–100 / 101–174 / 175–255)

Phase	VF % (his → app)	Mean µm (his → app)	σ (his → app)	µ (his → app)
NiO	43.0 → 43.0	1.08 → 0.99 (−8%)	0.79 → 0.84	−0.22 → −0.36
BZCY	55.8 → 55.8	1.31 → 1.28 (−2%)	0.87 → 0.91	−0.09 → −0.13

Phase	VF % (his → app)	Mean μm (his → app)	σ (his → app)	μ (his → app)
Pore	1.2 → 1.19	0.45 → 0.40 (−11%)	0.49 → 0.53	−0.92 → −1.08

OFA10 (bands 0–75 / 76–180 / 181–255)

Phase	VF % (his → app)	Mean μm (his → app)	σ (his → app)	μ (his → app)
NiO	52.4 → 51.3 (−1.1 pp)	1.42 → 1.32 (−7%)	0.83 → 0.87	0.03 → −0.07
BZCY	47.0 → 48.3 (+1.3 pp)	1.33 → 1.32 (−1%)	0.78 → 0.82	0.009 → −0.03
Pore	0.06 → 0.46	0.56 → 0.49 (−12%)	0.56 → 0.60	−0.72 → −0.90

OFA11 (bands 0–59 / 60–190 / 191–255)

Phase	VF % (his → app)	Mean μm (his → app)	σ (his → app)	μ (his → app)
NiO	57.0 → 57.0 (±0.0 pp)	0.97 → 0.91 (−6%)	0.71 → 0.75	−0.28 → −0.36
BZCY	41.6 → 41.6 (±0.0 pp)	0.91 → 0.85 (−7%)	0.64 → 0.70	−0.28 → −0.39
Pore	1.4 → 1.36	0.51 → 0.47 (−9%)	0.43 → 0.50	−0.76 → −0.89

3. Systematic bias (consistent across all three samples)

Quantity	Behaviour	Magnitude
Volume fraction	matches (threshold-driven, calibration-independent)	solids ⩽ ~1.3 pp; OFA11 exact
Mean diameter	app slightly **low**	−1% to −12%
Lognormal σ	app slightly **high**	+0.04 to +0.07 (pore up to +0.07)
Lognormal μ	app slightly low — follows from σ via $\mu = \ln(\text{mean}) - \sigma^2/2$	not an independent error

The direction and size of the bias are **stable across every sample and phase**. For the comparative OFA8/10/11 study this bias largely cancels, so relative rankings and trends are robust even though absolute σ carries a small positive offset.

Caveats (state in Limitations):

- OFA10 pore: app 0.46 % vs his 0.06 % — ~8× relative but only 0.4 pp absolute; the known narrow-pore-band segmentation sensitivity, not a pipeline defect.
- OFA11 pore σ runs +0.07 (+17%); pore is the smallest, noisiest phase throughout.
- 2D→3D reconstruction and watershed non-uniqueness on percolating microstructures remain core methodological limitations.

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4. Verdict

Quantity	OFA8	OFA10	OFA11	Status
Phase segmentation / VF	exact	≤1.3 pp	exact	**Validated**
Calibration (0.0365 μm/px)	✓	✓	✓	**Validated**
Mean diameter	−8%	−7%	−6%	**Validated (≤12%)**
Lognormal σ	+0.05	+0.04	+0.04	**Validated (was OPEN — now closed)**
Lognormal μ	consistent	consistent	consistent	Consistent (coupled to σ)

The σ item, previously bracketed uselessly between watershed-on (0.34) and watershed-off (0.94) modes on a non-matching crop, is **closed**: on the authentic images with regional-maxima seeding and matched thresholds, σ agrees within +0.04–0.07 on all three samples. The "report σ as a range" hedge can be dropped.

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5. Comparative study — validated app measurements (the science)

Side-by-side microstructure of the three electrodes, using the now-validated pipeline. These are the app's own numbers; OFA8 additionally carries the exact-VF MATLAB validation credential.

Quantity	OFA8 (sedimentary)	OFA10 (literature-preferred)	OFA11 (blistered)
Pore VF %	1.19	0.46	1.36
NiO VF %	43.0	51.3	57.0
BZCY VF %	55.8	48.3	41.6
NiO mean μm	0.99	1.32	0.91
BZCY mean μm	1.28	1.32	0.85
NiO σ	0.84	0.87	0.75
BZCY σ	0.91	0.82	0.70
NiO features	996	770	1749
BZCY features	772	763	1644

Preliminary microstructural signatures (to be tied to tortuosity + TPB density after the DREAM.3D 3D analysis): OFA11 has the finest NiO/BZCY grains and by far the most features; OFA10 has the coarsest solid grains and the most NiO-rich solid split; OFA8 sits between. These are the inputs to the performance-difference argument, not conclusions yet.

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6. Next steps

1. Run all three through Stage 4–5 (DREAM.3D synthesis → tortuosity + TPB density) at the shared 0.0365 μm/px.
2. σ-sensitivity check in DREAM.3D (generate at app-σ and his-σ; confirm the +0.04–0.07 offset doesn't move tortuosity/TPB rankings).
3. Fold this table set into the IMRaD Results/Validation section; cite his three threshold sets as evidence for per-sample thresholding.